THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The length of the diagonal of a square of side 8 cm, squared, is: 1

(a) 130 **(b)** 129 **(c)** 127 **(d)** 128

2. A man goes 30 m due east and then 40 m due north. His distance from the start (m) is: 1

(a) 52 **(b)** 49 **(c)** 51 **(d)** 50

3. In $\triangle ABC$, $AB = 10$, $BC = 11$, $AC = 15$. The triangle is: 1

(a) isosceles right **(b)** right-angled **(c)** equilateral **(d)** not right-angled

4. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 4 : 3$. Then $AE : AC$ is 4: 1

(a) 9 **(b)** 7 **(c)** 8 **(d)** 6

5. In $\triangle ABC$, $DE \parallel BC$, $AD = 6$ cm, $DB = 5$ cm, $AE = 12$ cm. Then EC (in cm) $=$ 1

(a) 9 **(b)** 11 **(c)** 12 **(d)** 10

6. $\triangle ABC \sim \triangle DEF$ with areas 9 cm² and 25 cm². If $AB = 9$ cm, then DE (cm) $=$ 1

(a) 15 **(b)** 14 **(c)** 16 **(d)** 17

7. The length of the diagonal of a square of side 9 cm, squared, is: 1

(a) 164 **(b)** 162 **(c)** 161 **(d)** 163

8. A man goes 21 m due east and then 28 m due north. His distance from the start (m) is: 1

(a) 36 **(b)** 35 **(c)** 37 **(d)** 34

9. In $\triangle ABC$, $AB = 20$, $BC = 48$, $AC = 52$. The triangle is: 1

(a) equilateral **(b)** isosceles right **(c)** right-angled **(d)** not right-angled

10. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 5 : 2$. Then $AE : AC$ is 5: 1

(a) 9 **(b)** 7 **(c)** 6 **(d)** 8

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The perimeters of two similar triangles are 15 cm and 60 cm. Find the ratio of their corresponding sides and their areas. 2

12. A ladder 41 m long reaches a window 40 m high. Find how far the foot of the ladder is from the wall. 2

13. Sides of a triangle are 9 cm, 10 cm, 14 cm. State with reason whether it is right-angled. 2

14. The perimeters of two similar triangles are 18 cm and 90 cm. Find the ratio of their corresponding sides and their areas. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 3$ cm, $DB = 8$ cm and $AE = 9$ cm, find EC. 3

16. The areas of two similar triangles are 12 cm² and 147 cm². If a side of the first is 2 cm, find the corresponding side of the second. 3

17. In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 2 : 5$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 7$ cm, $DB = 2$ cm and $AE = 14$ cm, find EC. 3

18. The areas of two similar triangles are 54 cm² and 96 cm². If a side of the first is 9 cm, find the corresponding side of the second. 3

OR

In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 3 : 2$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that if the square of one side of a triangle equals the sum of the squares of the other two sides, the angle opposite the first side is a right angle. 5

OR

Two similar triangles have perimeters 30 cm and 60 cm. If a side of the smaller is 8 cm, find the corresponding side of the larger and the ratio of their areas. 5

20. Prove that the line joining the midpoints of two sides of a triangle is parallel to the third side. 5

OR

Two similar triangles have perimeters 24 cm and 72 cm. If a side of the smaller is 7 cm, find the corresponding side of the larger and the ratio of their areas. 5

END OF QUESTION PAPER Total: 40 Marks

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